

Advanced Trees (Segment / Fenwick)

Range queries and updates efficiently.

Key concepts

- Segment tree: range sum/min/max in $O(\log n)$.
- Fenwick (BIT): prefix sums with less code.
- Great for range-update problems.

Key idea / example

Segment tree stores aggregates of ranges in a tree.

Today's tasks (1.5h)

- ☐ Understand segment tree idea
- ☐ Build a range-sum tree
- ☐ Compare with Fenwick

Resources

Playlist: youtube.com/playlist?list=PLsr8vTgyLdy_YndxNcI4WkH5Vorj5qvrV

(Clickable playlist link is in the app: Goals > this day > Study block.)